Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA (Remote ET)

PROFESSIONAL SUMMARY

Applied AI engineer who **builds production-grade agentic workflows end-to-end** - the exact work Cohere's FDE team ships for enterprise customers. Built AgentProbe: **production RAG + multi-agent system** (LangGraph, ReAct, Plan-and-Execute), **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama), and a **5-metric evaluation framework** with CI/CD gates - reliable, observable, auditable from day one. AI on FHIR: HIPAA-compliant clinical NLP platform for the **healthcare vertical**, one of Cohere's priority enterprise domains. **8+ years production Python**; enterprise-scale delivery at Entrata (100K+ MAU, 1.6M records, 99.5% uptime); M.S. Data Analytics Engineering, GMU (GPA 3.87). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Agentic Workflows (LangGraph, ReAct, Plan-Execute) | Production RAG Pipelines (ChromaDB) | LLM Evaluation Frameworks (LLM-as-Judge) | Multi-Provider LLM Routing | Prototype → Production (open-source) | Python (8+ years production) | Enterprise-Grade Reliability & Observability | Healthcare AI / HIPAA Compliance | Full-Stack End-to-End Ownership | FastAPI / REST APIs / Docker / AWS | Prompt & Context Engineering | Cross-Functional Technical Collaboration

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production AI on regulated data with output validation, compliance controls, and quality monitoring.
- Built and delivered **20+ full-stack modules** (Python, TypeScript/JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users** - owned use cases end-to-end from scoping to production, collaborating with PMs and business counterparts to translate requirements into working systems.
- Led Docker-based containerization for 30%+ of microservices, cutting execution time by **50%**; improved production query performance by **30%**; maintained **99.5% uptime** across enterprise SaaS.
- Migrated client data to **AWS S3** with Libsodium-encrypted pipelines - **100% PHI compliance, zero security incidents**; built 5+ platform tools adopted across squads, reducing resolution time by **40%**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Led technical discussions with 30+ students; translated complex concepts into clear specs and drove iterative quality improvement - same stakeholder communication pattern required when working with Cohere enterprise customers.

PROJECTS

AgentProbe: Production Agentic AI Platform - RAG, Multi-Agent, Evaluation OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

Built the same system Cohere's FDE team deploys for enterprise customers - production-grade agentic workflows from prototype to open-source. **Agentic workflows**: 3-agent LangGraph system (Orchestrator, RAG Tracer, Auto Fix) using ReAct and Plan-and-Execute patterns - multi-step reasoning, tool calls, failure handling, and inter-agent state handoffs. Agents reason, plan, and act across tools and data sources. **LLM stack**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task, latency, and capability; vector database (ChromaDB) + embeddings + retrieval quality scoring. **Evaluation framework**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with per-run regression tracking and CI/CD gates - agents are reliable, observable, and auditable from day one, exactly as Cohere requires. **Platform**: 15-endpoint FastAPI backend + React observability dashboard.
github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: Healthcare Enterprise AI - LLMs + FHIR APIs (HIPAA-Compliant)

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings

Production AI system for the healthcare enterprise vertical - one of Cohere's priority customer domains. Agentic platform routing natural language queries to FHIR APIs via LLMs, BERT, and embeddings. **HIPAA-compliant RBAC**, enterprise-grade access controls, regulated data. **~60% retrieval time reduction.** Proof of building AI for regulated enterprise workflows with safety and auditability requirements baked in from day one.
github.com/venkatalakshmiparimala/AI_on_FHIR

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation → deployment. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, Multi-Agent Orchestration, ReAct / Plan-and-Execute, LLM APIs (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Vector Search, LLM-as-Judge Evaluation, CI/CD Eval Gates, Prompt Engineering, Context Engineering

Backend & Infra: FastAPI, Flask, React, PostgreSQL, MySQL, Docker, AWS (S3, EC2, RDS, Glue), Azure, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & NLP: PyTorch, scikit-learn, BERT, spaCy, Hugging Face, NumPy, Pandas, Feature Engineering, Clinical NLP, FHIR APIs, HIPAA compliance

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum, Claude Code, GitHub Copilot